

Month	Core Focus Area	High-Impact Technical Achievements
January	Embedded Protocols & Research	• Researched core communication protocols including UART, I2C, and SPI.□ • Investigated sensor architectures, focusing on magnetometers and accelerometers.□ • Explored MicroPython environments and signal processing

		concepts via GNU Radio.
February	MicroPython & Kinematic Algorithms	- Developed baseline MicroPython firmware for GPIO and LED control in Thonny. - Engineered double-integration algorithms to derive velocity and displacement from accelerometer data. - Formulated spatial estimation models for tilt orientation and compass heading.

March	Hardware Schematic s & SOM Integratio n	• Designed custom schematic capture centered on the Logic PD System-on-Module (SOM).□• Built peripheral componen t libraries and establishe d bus interfaces across the PCB design.
April	Power Architectu re & Sensor Suites	• Architecte d the system power tree topology to optimize power distributio n.□•

		Integrated peripheral schematics including Audio Codec, IMU, Barometer, ALS, and E-Compass.
May	MCU Benchmarking & Power Selection	• Selected 5G mobile battery specifications based on operational power budgets. □ • Benchmarked microcontroller variants (RT1064 vs. Raspberry Pi Pico) to empirically validate and finalize

		the RT1064 design.
June	Signal Processing & GNSS Refinement	• Verified and revised GNSS Receiver schematics to eliminate layout defects and noise. □ • Implemented Kalman Filtering techniques to reduce sensor noise and smooth state estimates.
July	TinyML & Motion Analytics	• Deployed TinyML pipelines for real-time motion tracking

		and activity recognition. • Trained Decision Tree classifiers in Spyder using Scikit-learn. • Built data visualization pipelines using NumPy, Pandas, and Matplotlib to plot 2D/3D kinematic trajectories.
August	System Validation & Navigation	• Successfully validated and deployed an Indoor Navigation System

		functionin g reliably in GPS- denied environme nts.
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